Kinematics of Mechanisms

Homework 1

Report and Solutions

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1 Theoretical Questions

1.1 Question 1: Mechanism and machine

A *mechanism* is an arrangement of rigid bodies (links) connected by kinematic pairs so that their relative motion is constrained. Its purpose is to transmit or transform motion, such as converting a crank's continuous rotation into the oscillation of a rocker. A four-bar linkage is a mechanism.

A *machine* is an assembly that uses mechanisms together with other components to perform a useful task by transmitting or converting energy and forces. A reciprocating engine, for example, contains a slider-crank mechanism that converts piston motion into shaft rotation, while the complete engine also includes the energy source and supporting systems. Thus, "mechanism" emphasizes the prescribed relative motion; "machine" emphasizes useful function and power transfer. A machine may contain several mechanisms, and a mechanism can be part of a machine.

1.2 Question 2: Four-bar linkage and its types

A planar four-bar linkage is a closed chain of four rigid links joined by four revolute (pin) joints. One link is fixed as the frame; the other three are commonly called the input link, coupler, and output link. For an ordinary nondegenerate linkage, the planar mobility equation gives

$$M = 3(n-1) - 2J_1 - J_2 = 3(4-1) - 2(4) - 0 = 1, \tag{1}$$

where n includes the frame, J_1 counts one-DOF joints, and J_2 counts two-DOF joints. One independent input therefore fixes the configuration within a chosen assembly branch.

The usual types describe how the two links attached to the frame move:

- **Double-crank (drag link):** both adjacent links can make full revolutions relative to the frame.
- **Crank-rocker:** one adjacent link can make a full revolution; the other only oscillates through a bounded angle.
- **Double-rocker:** neither adjacent link can make a full revolution; both oscillate.

Which type results depends on both the link lengths and the choice of fixed link. The coupler's motion is not itself used to name these three types. The strict Grashof classifications are given in Question 4.

1.3 Question 3: Main lower kinematic pairs and their DOF

A lower kinematic pair connects two bodies through contacting surfaces. Its DOF is the number of independent *relative* motion coordinates allowed by the joint. In three-dimensional space, the common lower pairs are as follows.

Table 1: Main lower pairs and relative spatial mobility.

Pair	DOF	Permitted relative motion
Revolute (R)	1	Rotation about a fixed axis.
Prismatic (P)	1	Translation along a fixed axis.
Helical (H)	1	Coupled rotation and translation along an axis, with a fixed screw pitch.
Cylindrical (C)	2	Independent rotation about and translation along the same axis.
Spherical (S)	3	Rotation about three independent axes through a common center; no translation.
Planar (E)	3	Two independent translations in a plane and one rotation normal to it.

The values in Table 1 are *spatial* joint DOFs. For the planar mobility count in Part A, a planar revolute or prismatic pair is a one-DOF joint, whereas a general planar higher pair permits two relative DOFs. Joint DOF must not be confused with the mobility of the complete mechanism.

1.4 Question 4: Grashof, non-Grashof, and special case

Let S and L be the shortest and longest lengths of a four-bar chain, and let P and Q be the other two lengths. For a nondegenerate four-bar, Grashof's criterion compares the two sums:

$$\begin{aligned} S+L &< P+Q \quad \text{strict Grashof chain,} \\ S+L &> P+Q \quad \text{non-Grashof chain,} \\ S+L &= P+Q \quad \text{Grashof equality, or change-point case.} \end{aligned} \tag{2}$$

For a **strict Grashof chain**, at least the shortest link can rotate fully relative to another link. Fixing the shortest link gives a double-crank; fixing a link adjacent to the shortest gives a crank-rocker; fixing the link opposite the shortest gives a double-rocker. These statements assume distinct ordinary link lengths and a feasible assembly.

For a **non-Grashof chain**, no link can rotate through a complete revolution relative to the other links. With any link fixed, the ordinary inversion is a double-rocker.

At **equality**, $S + L = P + Q$, the linkage can reach a collinear *change-point* configuration. Its motion there is singular: the usual classification becomes a limiting case, and continuation through the aligned position can depend on the assembly branch or physical guidance. It should therefore be reported separately instead of being treated as an ordinary strict Grashof or non-Grashof linkage.

2 Part A: Cinderella Models and Mobility

Create each mechanism in Cinderella, label its mobility with the `Add text` tool, and insert a screenshot below. Submit the three corresponding `.cdy` files with the report. For each model, identify *rigid bodies*, including the ground, before counting joints. If more than two links meet at one pin, count the equivalent binary joints appropriately. Explain any pin-in-slot or contact modeling convention.

For a general planar mechanism, use the Grübler–Kutzbach count

$$M = 3(n-1) - 2J_1 - J_2, \tag{3}$$

where n is the number of links including ground, J_1 the number of one-DOF lower pairs, and J_2 the number of two-DOF higher pairs. Check the count against observed independent motions, especially if a mechanism has redundant constraints or a special geometry.

2.1 Case 1: Compound linkage

The dimensions supplied in the assignment are

$$AB = 2,\ BC = 9,\ CD = 6,\ DE = 6,\ EQ = 16,\ QG = 6,$$
$$EF = 8,\ QL = 8,\ DP = 9,\ PG = 7,\ PA = 4,$$
$$\angle DEF = 15^\circ, \qquad \angle LQG = 15^\circ.$$

[To complete: Insert Cinderella screenshot: `figures/part_a_case1.png`]

Show the assembled mechanism and its DOF annotation.

Figure 1: Case 1 Cinderella model with its DOF annotation.

[To complete: List the rigid links and identify all pin joints, including any compound joints.] The count is

$$n = _____, \qquad J_1 = _____, \qquad J_2 = _____,$$
$$M = 3(n-1) - 2J_1 - J_2 = _____.$$

[To complete: Explain any special geometric or redundant constraints and verify the observed mobility in Cinderella.]

2.2 Case 2: Casement window mechanism

Use $AB = 4$, $BC = 8$, $CD = 5$, $DE = 7$, $EF = 5$, with the fixed angles $\theta = 125^\circ$ and $\varphi = 125^\circ$. Points D and F are sliders.

[To complete: Identify each rigid link, each revolute pair, and each prismatic pair; explain how the fixed angles are represented.]

$$n = _____, \qquad J_1 = _____, \qquad J_2 = _____,$$
$$M = 3(n-1) - 2J_1 - J_2 = _____.$$

[To complete: Verify the observed mobility in Cinderella.]

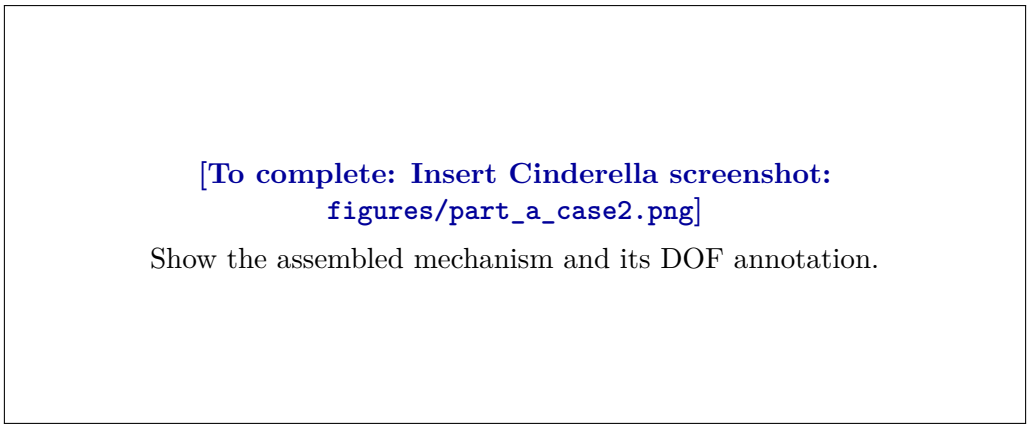


Figure 2: Case 2 Cinderella model with both sliders and its DOF annotation.

2.3 Case 3: Bull gear and slotted bar

Use $AC = 4$, $AB = 8$, $AP = 8$, and $KB = 20$.

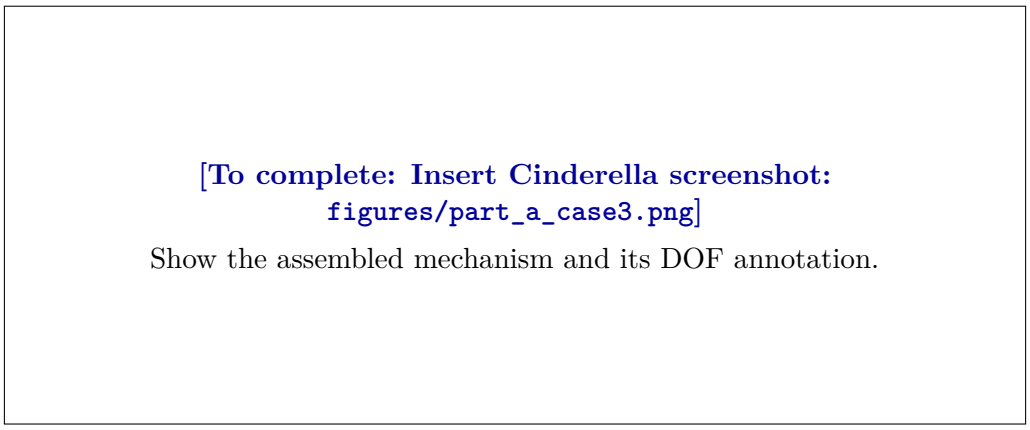


Figure 3: Case 3 Cinderella model, including gear, slot, slider, and DOF annotation.

[To complete: Describe the constraints at the gear, connecting rod, slot, and horizontal slider. State whether a pin-in-slot is counted as a higher pair or modeled with an equivalent slider and lower pairs; use one consistent count.]

$$n = _____, \qquad J_1 = _____, \qquad J_2 = _____,$$
$$M = 3(n-1) - 2J_1 - J_2 = _____.$$

[To complete: Verify the observed mobility in Cinderella.]

Table 2: Part A mobility results.

Mechanism	Links n	One-DOF joints J_1	Two-DOF joints J_2	Mobility M
Case 1				
Case 2				
Case 3				

3 Part B: Grashof Condition and Linkage Classification

3.1 Task 1: Classify three four-bar chains

All lengths in Table 3 are in centimetres. In each case, sort the four lengths into S, L, P, Q, compare $S + L$ with $P + Q$, and name the mechanism types obtainable by selecting different fixed links.

Table 3: Given four-bar link lengths in centimetres.

Case	l_1	l_2	l_3	l_4
1	12	4	10	9
2	4	8	11	7
3	6	3	5	4

Table 4: Part B, Task 1 working table.

Case	S	L	P, Q	$S + L$	$P + Q$	Class and possible fixed-link types
1	____	____	____	____	____	**[To complete: Class and types]**
2	____	____	____	____	____	**[To complete: Class and types]**
3	____	____	____	____	____	**[To complete: Class and types]**

[To complete: For each case, show the inequality or equality explicitly. State which link is fixed for every claimed double-crank, crank-rocker, or double-rocker inversion. Treat a change-point case separately.]

3.2 Task 2: Design a double-crank mechanism

The fixed link is $l_1 = 4\,\text{cm}$, and the input link is $l_2 = 9\,\text{cm}$. Choose integer coupler and output lengths.

$$l_3 = _____\,\text{cm}, \qquad l_4 = _____\,\text{cm}.$$

[To complete: Identify S, L, P, Q; demonstrate the strict inequality $S + L < P + Q$; confirm that the fixed link l_1 is the shortest.]

4 Part C: Numerical Position Analysis of a Planar Four-Bar

Take $A^* = (0, 0)$ and $B^* = (r_1, 0)$. The input link runs from A^* to A, the coupler from A to B, and the output link from B^* to B. Angles $\theta_2, \theta_3, \theta_4$ are measured counterclockwise from the positive x-axis. The loop closure is

$$r_2 e^{i\theta_2} + r_3 e^{i\theta_3} = r_1 + r_4 e^{i\theta_4}. \tag{4}$$

For numerical solution, use the two residual equations

$$f_1(\theta_3, \theta_4) = r_2 \cos\theta_2 + r_3 \cos\theta_3 - r_1 - r_4 \cos\theta_4 = 0, \tag{5}$$

$$f_2(\theta_3, \theta_4) = r_2 \sin\theta_2 + r_3 \sin\theta_3 - r_4 \sin\theta_4 = 0. \tag{6}$$

One suitable numerical method is Newton–Raphson iteration,

$$\begin{bmatrix}\theta_3\\ \theta_4\end{bmatrix}_{k+1} = \begin{bmatrix}\theta_3\\ \theta_4\end{bmatrix}_{k} - J(\boldsymbol{\theta}_k)^{-1}\begin{bmatrix}f_1\\ f_2\end{bmatrix}_{k}, \qquad J = \begin{bmatrix}-r_3\sin\theta_3 & r_4\sin\theta_4\\ r_3\cos\theta_3 & -r_4\cos\theta_4\end{bmatrix}. \tag{7}$$

Use radians during evaluation and convert the final angles to degrees. State the initial guess and the selected assembly branch (open or crossed), because two configurations may satisfy the same input data. After finding the angles, compute the required diagonal $L = |A^*B|$ from

$$L = \sqrt{(r_1 + r_4\cos\theta_4)^2 + (r_4\sin\theta_4)^2}. \tag{8}$$

Report a small final residual as a check of the numerical solution.

4.1 Task 1

Given $r_1 = 90$, $r_2 = 40$, $r_3 = 140$, $r_4 = 110$, and $\theta_2 = 55^\circ$ (length units as supplied in the assignment).

Initial guess (θ_3, θ_4) = **[To complete: angles and branch]**,
Numerical method and stopping criterion = **[To complete: method and tolerance]**,
$\theta_3 =$ _____°, $\theta_4 =$ _____°,
$L = |A^*B| =$ _____, $\|(f_1, f_2)\| =$ _____.

[To complete: Show the substituted residual equations and enough iteration or solver detail to make the numerical result reproducible.]

4.2 Task 2

Given $r_1 = 70$, $r_2 = 55$, $r_3 = 120$, $r_4 = 95$, and $\theta_2 = 115^\circ$ (length units as supplied in the assignment).

Initial guess (θ_3, θ_4) = **[To complete: angles and branch]**,
Numerical method and stopping criterion = **[To complete: method and tolerance]**,
$\theta_3 =$ _____°, $\theta_4 =$ _____°,
$L = |A^*B| =$ _____, $\|(f_1, f_2)\| =$ _____.

[To complete: Show the substituted residual equations and enough iteration or solver detail to make the numerical result reproducible.]

5 Summary of Results

The compact table below follows the grouping in the assignment's example. Give the essential result in each cell; the detailed counts and calculations remain in Parts A–C.

Table 5: Answer summary for all tasks.

Part A: mobility	Part B: classification and design	Part C: position
Case 1: $M =$ ____	Task 1, Case 1: ____	Task 1: $L =$ ____, $\theta_3 =$ ____°, $\theta_4 =$ ____°
Case 2: $M =$ ____	Task 1, Case 2: ____	Task 2: $L =$ ____, $\theta_3 =$ ____°, $\theta_4 =$ ____°
Case 3: $M =$ ____	Task 1, Case 3: ____	
	Task 2: $l_3 =$ ____ cm, $l_4 =$ ____ cm	